

Dual-theory reformulation of hydrogen-like precision spectroscopy

Trey Morris with Claude Opus 4.7

2026-05-28

Dual-theory reformulation of hydrogen-like precision spectroscopy — intro note for a new collaborator

Date: 2026-05-28. **From:** Trey Morris (with Claude Opus 4.7). **Re:** Draft introductory email to a new experimental collaborator in optical/microwave atomic spectroscopy. Tracks against [issue #88](#). Methods-focused framing; honest-scope paragraph is load-bearing and must be retained before sending. Source material: the Bethe–Salpeter precision campaign ([#50](#)), the Li^{2+} Z-extension ([#78](#)), and the hydrogenic-ion g-factor Z-scan ([#82](#)).

Draft email

Replace [Name], the salutation, and the sign-off before sending. Verify the numbers against the provenance table below.

Subject: Dual-theory reformulation of hydrogen-like precision spectroscopy — would value your read

Dear [Name],

I’m a co-author on a reformulation of relativistic quantum mechanics — Gill’s “dual theory,” built on a proper-time formulation — and we’ve been working through its predictions for precision atomic spectroscopy. Since your work is in optical/microwave spectroscopy of hydrogen-like systems, I’d value your read on the apparatus and your sense of which measurements are most diagnostic. This note summarizes the method and where it stands against measurement.

The method, in one paragraph. The dual theory recasts relativistic dynamics through a proper-time canonical Hamiltonian, $K = H^2/(2mc^2) + mc^2/2$, with a dual-Dirac equation in place of the standard Dirac equation. For precision atomic observables this reduces to a clean structure: each g_s -dependent observable is the textbook (Sommerfeld–Dirac / Fermi-contact) result times $(g_s/-2)^n$, with $n = 1$ for spin–orbit and Fermi-contact contributions and $n = 2$ for two-fermion spin–spin couplings. The electron spin g-factor g_s itself enters through a single cutoff parameter in the framework’s §III.D structure, and the whole apparatus scales across the hydrogen-like sequence with the expected Z-powers. The upshot is that one consistent prescription handles the Lamb shift, fine structure, and hyperfine splitting across H, He^+ , Li^{2+} , and up the isoelectronic sequence.

Where it stands against measurement (optical/microwave regime).

Observable	Measurement	Framework	Note
H Lamb shift $2S_{1/2}-2P_{1/2}$	1057.845(9) MHz	~ 1016 MHz	Bethe-1947 estimate; ~ 42 MHz ($\sim 4\%$) residual at the estimate's precision floor
H fine structure $2P_{3/2}-2P_{1/2}$	10 969.13(10) MHz	10 962 MHz	leading Dirac x anomalous- g
H 1S hyperfine (21-cm)	1 420.405 751 768(2) MHz	1 420.04 MHz	Fermi contact at the framework cutoff; ~ 0.4 MHz residual
Li^{2+} Lamb shift $2S_{1/2}-2P_{1/2}$	62 765(21) MHz (Schiffer 1995)	$\sim 61-63$ GHz	Z^4 -scaled Bethe-estimate
Li^{2+} fine structure $2P_{3/2}-2P_{1/2}$	— (no direct precision measurement located)	≈ 888 GHz	$Z^4 \times \text{H}$
Li^{2+} 1s hyperfine	— (theory comparator: Pachucki 2023)	≈ 29.8 GHz	$Z^3 \times$ nuclear factors, $I = 3/2$

Honest scope. At the precision the apparatus currently delivers — the leading- g_s / Bethe-estimate floor — the framework reproduces the standard QED results rather than predicting deviations from them; the agreement is by construction of the reformulation, not an independent corroboration. The frontier where a genuinely distinct prediction could appear is a full proper-time one-loop calculation, which we have not yet completed. I mention this up front because I'd rather be precise about what's reformulation and what would be new physics.

What would be most useful from you. Two things: (1) an expert read on whether the proper-time / dual-Dirac reduction is faithful to how these observables are actually measured and extracted; and (2) your sense of which optical/microwave measurements — higher- Z hydrogen-like fine structure or hyperfine, improved $2S-2P$ intervals, or something else — would most sharply test a reformulation of this kind, particularly any regime where the proper-time radiation-reaction structure might leave a measurable signature.

Happy to send the full per-observable derivations (each reproduced symbolically in Mathematica) if useful. Thank you for considering it.

Best regards, [Trey Morris]

Backing provenance table (for Trey to verify before sending — not part of the email)

Observable	Framework value	Source in repo	Measurement	Measurement source
H Lamb shift $2S_{1/2}-2P_{1/2}$	~ 1016 MHz	Bethe_Salpeter/10_CrossComparison.md PR E row	$1057.845(9)$ MHz	CODATA-2018
H fine structure $2P_{3/2}-2P_{1/2}$	$10\,962$ MHz	03_FineStructure.md BS-§14.2	$10\,969.13(10)$ MHz	Hagley & Pipkin 1994
H 1S hyperfine (21-cm)	$1\,420.04$ MHz	10_CrossComparison.md §1 residual table	$1420.405\,751\,768(2)$ MHz	NIST / Karshenboim review
Li^{2+} Lamb shift	$\sim 61\text{--}63$ GHz	12_Li2plus_Spectroscopy.md §12.2 (branch #85)	$63\,765(21)$ MHz	Schiffer et al., <i>PRL</i> 74 , 2188 (1995)
Li^{2+} fine structure	$887,920$ MHz	12_Li2plus_Spectroscopy.md §12.3 (branch #85)	not added	— (theory: Z^4 -Dirac)
Li^{2+} 1s hyperfine	~ 29.8 GHz	13_Li2plus_Hyperfine.md (branch #86)	not added	Pachucki et al. 2023 (theory from Li^+)

Note on the Li^{2+} rows: the Li^{2+} predictions live on the not-yet-merged PR branches [#85](#) (Lamb shift + fine structure) and [#86](#) (hyperfine). Verify against the merged versions once those PRs land, or against the branch docs in the interim. The Li^{2+} fine-structure and hyperfine rows have **no direct precision measurement** — the email marks them “—” honestly; do not imply an experimental comparison exists for those two.

Note on the honest-scope paragraph: it is consistent with the campaign’s own framing in [10_CrossComparison.md](#): “zero of the 28 recorded results constitute independent corroborations of the dual theory’s content distinct from textbook QM.” The hydrogenic-ion g-factor Z-scan ([#82](#)) independently confirmed this on the Z-axis (the framework’s cutoff is Z-trivial; it inherits QED’s bound-state structure rather than predicting deviations). Keeping the honest-scope paragraph is non-negotiable for an external-scientist audience.